



The Central Limit Theorem · Worksheet 1

Further Statistics 1 Chapter 5: The Central Limit Theorem. Answer all questions. Show your working clearly. Total: 68 marks. Time: 85 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are three sections, one for each topic in Chapter 5. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Whenever you use the Central Limit Theorem, say so and state why it applies (e.g. “ n is large”). Remember that $\bar{X}$ has **variance** $\frac{\sigma^2}{n}$. Give probabilities to 4 decimal places.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A The distribution of the sample mean	21	
B Sample totals	21	
C Applying the CLT to other distributions (and when it is needed)	21	
Roar-level challenge	5	
Total	68	

Section A The distribution of the sample mean

21 marks

1. The random variable X has mean 12 and variance 20. The distribution of X is not normal. A random sample of 40 observations of X is taken. [2]

Write down, with a reason, the approximate distribution of the sample mean $\bar{X}$.

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2. The time taken by a call-centre operator to answer a call has mean 45 seconds and standard deviation 12 seconds. The distribution of times is positively skewed. A random sample of 36 calls is taken. Find the approximate probability that the mean answering time of the sample is less than 43 seconds. [3]

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3. A random sample of 50 observations is taken from a population with mean 8 and variance 18. Find, approximately, $P(7.5 < \bar{X} < 9)$. [3]

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4. A population has unknown mean μ and standard deviation 5. A random sample of size n is taken. Using a suitable approximation, find the smallest value of n for which $P(|\bar{X} - \mu| < 1) \geq 0.95$. [4]

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5. The discrete random variable X has the probability distribution below.

x	0	1	4
$P(X = x)$	0.4	0.4	0.2

(a) Show that $E(X) = 1.2$ and find $\text{Var}(X)$. [3]

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(b) A random sample of 54 observations of X is taken. Find an estimate of the probability that the sample mean is greater than 1.5. [3]

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(c) Repeat part (b) for a random sample of 96 observations, and comment on the effect of increasing the sample size. [2]

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(d) Explain why the Central Limit Theorem is needed in part (b). [1]

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(Total for Question 5 is 9 marks)



Section B Sample totals

21 marks

6. The random variable X has mean 5 and variance 4. The random variable T is the sum of a random sample of 50 observations of X . [2]
Write down the approximate distribution of T .

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7. The masses of apples have mean 120 g and standard deviation 15 g. A bag contains 40 randomly chosen apples. [3]
Find the approximate probability that the total mass of the apples in the bag is more than 4900 g.

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8. The masses of cars using a ferry have mean 1.4 tonnes and standard deviation 0.3 tonnes. The ferry carries 50 cars, chosen at random, and its safe load for cars is 72 tonnes. [3]
Find the approximate probability that the total mass of the 50 cars is more than the safe load.

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9. The time taken to serve a customer at a counter has mean 3 minutes and standard deviation 2 minutes, independently for each customer. The counter is open for 480 minutes each day. [4]
Find the largest number of customers, n , that can be booked in so that the probability that all of them are served within the 480 minutes is at least 0.95.

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10. The masses of a brand of cereal bar have mean 32 g and standard deviation 1.5 g. The bars are sold in packs of 48 randomly chosen bars. The random variable T is the total mass of the bars in a pack.

(a) Write down, stating the name of the theorem you use, the approximate distribution of T . [2]

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(b) Find $P(T > 1540)$. [3]

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(c) The manufacturer changes the mean mass of a bar to μ g, leaving the standard deviation unchanged, so that only 1% of packs have a total mass of less than 1520 g. Find the value of μ . [3]

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(d) Explain why an approximation is needed in part (b). [1]

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(Total for Question 10 is 9 marks)

**Section C Applying the CLT to other distributions**

21 marks

11. For each of the following, state whether the Central Limit Theorem is needed to find the distribution of the sample mean. Give a reason. [2]

(a) A random sample of 5 observations from $N(10, 9)$.

(b) A random sample of 40 observations from $Po(3)$.

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12. A random sample of 50 observations is taken from a Poisson distribution with mean 4. [3]
Find the approximate probability that the sample mean is greater than 4.5.

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13. The random variable $X \sim B(10, 0.3)$. A random sample of 35 observations of X is taken. [3]
Find the approximate probability that the sample mean is less than 2.8.

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14. A fair nine-sided spinner has sides labelled 1, 2, 3, ..., 9. The score on one spin is X , so that X has a discrete uniform distribution. The spinner is spun 30 times. [4]
Find the approximate probability that the mean score is between 4.5 and 5.5.

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15. A game is played repeatedly until a player wins. The number of games played, X , up to and including the first win has a geometric distribution with parameter 0.25.

(a) Write down the mean and the variance of X . [2]

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(b) A random sample of 48 values of X is taken. Find the approximate probability that the sample mean is greater than 4.8. [3]

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(c) Find the smallest sample size n for which the approximate probability that the sample mean is greater than 4.8 is less than 0.01. [3]

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(d) Explain why the sample mean does not have an exact normal distribution. [1]

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(Total for Question 15 is 9 marks)



ROAR-LEVEL CHALLENGE

5 marks

16. A fair six-sided die is rolled n times and the mean score, $\bar{X}$, is found. [5]
Using a suitable approximation, find the smallest value of n for which $P(|\bar{X} - 3.5| < 0.1) \geq 0.9$.

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END OF WORKSHEET

Total: 68 marks